# FRANCES (SU) LIU

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## RESEARCH INTERESTS

Financial machine learning; financial econometrics and forecasting; high-frequency and alternative data; portfolio construction and risk modeling.

My research studies when richer information sets and more flexible models improve financial forecasts, and how those gains vary across market states. I am extending this work toward high-dimensional portfolio allocation and empirical asset pricing, where the same question governs whether added model complexity earns its place.

## EDUCATION

**Doctor of Philosophy (PhD) in Finance** *2024.02–2027.08 (expected)*
*University of Technology Sydney (UTS), Sydney, Australia*

*Supervisors*: Assoc. Prof. Vitali Alexeev (UTS, Finance), Assoc. Prof. Katja Ignatieva (UNSW, Actuarial) & Prof. Adam Clements (QUT, Finance)

**Master of Science in Financial Engineering** *2015.08–2017.06*
*Stevens Institute of Technology, New Jersey, United States* *GPA: 3.9/4.0*

*Thesis*: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence across financial sectors, time-series CoVaR). Supervised by Assoc. Prof. Zhenyu Cui

**Bachelor of Economics in Financial Engineering** *2011.09–2015.06*
*Harbin University of Commerce, Harbin, China*

## RESEARCH PAPERS

**Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting**
with Vitali Alexeev (UTS), Adam Clements (QUT), and Katja Ignatieva (UNSW) *working paper*

- Replaces the coarse overnight aggregates used in the volatility literature with the complete tick-level record of each extended-hours session, encoding variable-length, irregularly spaced transaction sequences and integrating the learned representations with a HAR benchmark
- Develops an incremental architecture hierarchy separating the contributions of non-linearity, tick-level information, trade weighting, and sequence ordering, which distinguishes information gains from gains attributable to model capacity
- Applied to all DJIA constituents from 2002 to 2025, reduces one-day-ahead out-of-sample forecast loss by approximately 11%, roughly twice the gain of the strongest scalar overnight proxy, with improvements stable across the Global Financial Crisis and COVID-19
- Finds that pre-open and post-close sessions generate comparable aggregate gains through different mechanisms: 64% of the pre-open improvement is recoverable from a within-session aggregate, while 81% of the post-close improvement is attributable to tick-sequence structure
- *Accepted*: Econometric Society Australasia Meeting (Adelaide, November 2026)
- *JEL*: C45, C53, G12, G14

**When Does Sentiment Predict Volatility?**
with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW) *working paper*

- Evaluates whether vendor-derived Thomson Reuters MarketPsych sentiment indices add predictive information to HAR volatility forecasts, and shows that modest average gains conceal substantial state dependence
- Finds that sentiment becomes more informative in the upper tail of the volatility distribution and when volatility persistence weakens, with persistence measured in a fractional-integration framework
- Shows that out-of-sample gains concentrate in jointly high-volatility and low-persistence regimes, consistent with sentiment supplying forward-looking narrative information when price-based predictors are least reliable

- *Presented*: International Association for Applied Econometrics (IAAE) Annual Conference (Carcavelos, June 2026)
- *Accepted*: Econometric Society Australasia Meeting (Adelaide, November 2026); FIRN Annual Conference (K'gari, November 2026)
- *JEL*: C22, C53, G12, G14

## AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend) 2024–2027
UTS International Research Scholarship (full tuition) 2024–2027
Academic Excellence Award, Stevens Institute of Technology 2017

## TEACHING EXPERIENCE

**University of Technology Sydney** *Sydney, Australia*
*Tutor, Finance Discipline Group* 2024–present

- 25503 *Investment Analysis* (undergraduate), Spring 2026
- 25556 *The Financial System* (undergraduate), Autumn and Spring 2026

## INDUSTRY EXPERIENCE

Six years of applied quantitative research across systematic strategies, portfolio construction and asset allocation, risk modeling, structured-finance valuation, and research infrastructure, carrying models from hypothesis through backtesting to deployment.

**Webull Financial LLC** *Changsha, China*
*Senior Quantitative Analyst* 2020.04–2023.11

- Developed portfolio construction and asset allocation for a client-facing investment-advisory product spanning five risk profiles, combining screened ETF universes, mean-variance optimization, CAPM-implied returns, machine learning denoising, and covariance shrinkage
- Researched and implemented mid-frequency strategies across Chinese and US equity markets, including factor-based stock selection, sector allocation, ETF rotation, and equity-index strategies
- Took strategies from research and backtesting through paper trading to live deployment, with ongoing post-deployment performance monitoring, using S&P, FactSet, and Refinitiv data
- Designed and co-developed the shared research database and backtesting environment used by the 12-person private-fund quantitative research team

**Pricing and Analytics (Shenzhen) Limited** *Shenzhen, China*
*Quantitative Analyst: Derivatives Pricing and Valuation* 2017.12–2019.10

- Developed dynamic pricing, valuation, and risk models for property-backed REITs and CLOs issued by listed real-estate companies across successive issues; two models were registered as software copyrights
- Modeled underlying cash flows and calibrated prepayment assumptions from market-observable spreads, and applied Black-Scholes, risk-neutral and martingale pricing, extreme value theory, and Value at Risk for portfolio tail risk
- Implemented the pricing engines in C++ with Excel interfaces and VBA integration, with relative valuation against market comparables selected at city and listed-issuer level

## COMPUTING AND LANGUAGES

**Computing:** Python (PyTorch, pandas, NumPy), GPU/HPC cluster computing, SQL, R, C++
**Languages:** Mandarin (native), Cantonese (fluent), English (fluent)